

10 — AI & Machine Learning Algorithms

■ AI/ML Implementation Status Summary

- **AI/ML Status: FULLY IMPLEMENTED & GROUNDED IN CODE**
- **Primary Source Files:**
- ***CODE_PH_0*** (Holt-Winters Exponential Smoothing + Ridge Regression Forecasting & WAPE Backtesting Engine).
- ***CODE_PH_0*** (Isolation Forest, LSTM, Autoencoder, Transformer anomaly models & Monte Carlo simulator).
- ***CODE_PH_0*** (LangGraph 3-Tier Cognitive Query Dispatcher).

■ 1. Primary Time-Series ML Forecasting Engine

Model Architecture & Math

The primary forecasting engine combines **Holt-Winters Exponential Smoothing** (to capture seasonal cycles and level trends) with **Multi-Feature Ridge Regression with L2 Regularization** (*CODE PH 0*).

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##### ■ Time-Series Feature Engineering ■ ■ -
Lags: y(t-1), y(t-12) ■ ■ - Rolling 3-month moving average ■ ■ - Vessel arrival counts ■ ■ - Seasonal
month indicator dummies ■ ■ ##### ■ ▼
##### ■ Dual ML Model Execution ■ ■ 1.
Holt-Winters Exponential Smoothing ■ ■ 2. Multi-Feature Ridge Regression (L2) ■
##### ■ ▼
##### ■ 95% Dynamic Prediction Intervals ■ ■
Upper = Forecast + 1.96 * StdErr * (1 + 0.05*t) ■ ■ Lower = Forecast - 1.96 * StdErr * (1 + 0.05*t) ■
#####

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Input Features & Preprocessing

- **Input Dataset:** Historical monthly cargo tonnage by commodity from ***CODE_PH_0***.
- **Engineered Features:**
 1. ***CODE_PH_0***: Tonnage in the previous month (\$t-1\$).
 2. ***CODE_PH_0***: Tonnage in the same month of the previous year (\$t-12\$).
 3. ***CODE_PH_0***: 3-month rolling average volume.
 4. ***CODE_PH_0***: Number of ship calls per commodity.
 5. ***CODE_PH_0*, *CODE_PH_1***: Cyclical sine/cosine month encoders.

■ 2. Model Evaluation & Out-of-Sample Backtesting

To ensure models are mathematically verified rather than arbitrary lines, the system executes chronological out-of-sample backtesting comparing a **Baseline Model (Seasonal Naive)** against the **Primary ML Model (CODE_PH_0)**.

Evaluation Metrics Calculated:

1. WAPE (Weighted Absolute Percentage Error) — Primary Recommended Metric:

$$\text{WAPE} = \frac{\sum |y_t - \hat{y}_t|}{\sum y_t} \times 100$$

Why WAPE?: Prevents division-by-zero errors when monthly commodity traffic drops to zero tonnes.

2. MAPE (Mean Absolute Percentage Error):

$$\text{MAPE} = \frac{1}{n} \sum \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100$$

3. MAE (Mean Absolute Error in Tonnes):

$$\text{MAE} = \frac{1}{n} \sum |y_t - \hat{y}_t|$$

4. RMSE (Root Mean Squared Error in Tonnes):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (y_t - \hat{y}_t)^2}$$

5. Model Accuracy Score:

$$\text{Accuracy \%} = 100 - \text{WAPE}$$

3. Quantified Forecast Driver Explainability

The system decomposes predictions into transparent, percentage-weighted driver components (CODE_PH_0):

- **Hinterland Industrial Demand:** Weight 35–45% (MRPL Refinery crude input, UPCL thermal coal requirements).
- **Monsoon Swell Factor:** Weight 20–25% (Seasonal wave swells impacting unloading speeds).
- **Trade Lane Momentum:** Weight 15–20% (YoY trade velocity and commodity price indices).

4. Probabilistic & Anomaly Models

1. Interactive What-If Simulator (CODE_PH_0):

- Adjusts vessel arrival %, trade demand %, and weather delay days to calculate net throughput deltas and capacity stress levels (CODE_PH_0, CODE_PH_1, CODE_PH_2).

2. Monte Carlo Policy Simulator (CODE_PH_0):

- Runs 1,000 statistical iterations over handling charge and tariff variations to generate P10, P50, and P90 revenue quantiles.

3. Multi-Algorithm Anomaly Engine (CODE_PH_0):

- Simulates Isolation Forest (outliers), LSTM (sequence deviations), Autoencoder (reconstruction error), and Transformer models to classify cargo volume surges, volume decline anomalies, and weather disruptions into severity tiers

(CODE_PH_0).